# Block practice · Rare extras

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Explain how mosquitoes and houseflies spread disease

Explain how enzymes work in biosensors and test strips

Explain how the pupil responds to changes in light intensity

Interpret a menstrual-cycle graph of hormones or lining thickness

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# Practice workbook · Explain how mosquitoes and houseflies spread disease

Mosquitoes carrying malaria parasites and houseflies carrying bacteria, and reasons disease levels differ between places.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-VECTOR-TRANSMISSION-DISTRIBUTION

# 1. Question 8 · part a · 3 marks

June 2019 Paper 1

Worked-example retry · Attempt part a only.

**8** *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

**(3)**

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

It shows the data for five different regions of the world in 2015.

| **Region of the world** | Q | R | S | T | U |
|---|---|---|---|---|---|
| **Total number of deaths from malaria** | 191 000 | 800 | 3800 | 14 400 | 1200 |
| **Number of deaths caused by *P. vivax*** | 1000 | 500 | 1400 | 4900 | 700 |

(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

**(3)**

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## 1. Question 8 · part a · 3 marks · Working and answer

Working and answer

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## 2. Question 4 · part b.i · 1 mark

January 2023 Paper 1

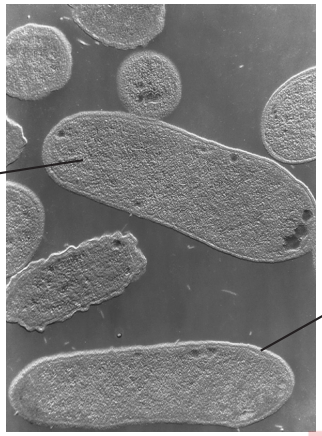
Further practice · Attempt part b.i only.

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**4** Typhoid is caused by *Salmonella typhi* bacteria.

These bacteria are spread by houseflies.

(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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## 2. Question 4 · part b.i · 1 mark

January 2023 Paper 1

Further practice · Attempt part b.i only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

**(1)**

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

**(3)**

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## 2. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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## 3. Question 4 · part b.ii · 3 marks

January 2023 Paper 1

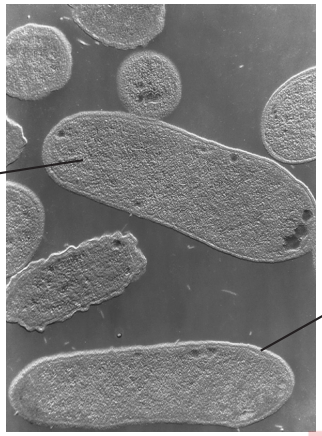
Further practice · Attempt part b.ii only.

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(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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## 3. Question 4 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

**(1)**

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

**(3)**

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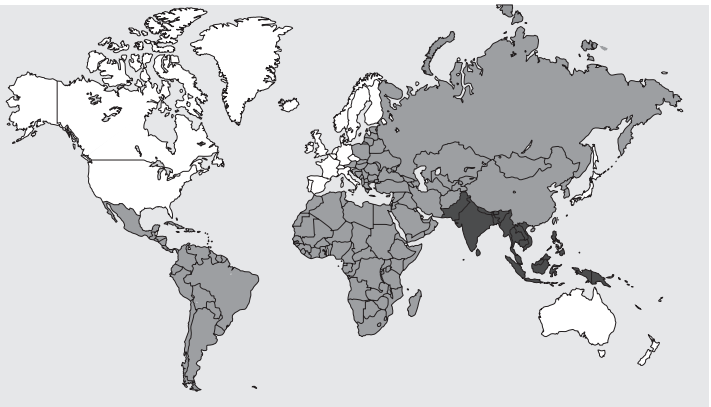
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## 3. Question 4 · part b.ii · 3 marks

January 2023 Paper 1

Further practice · Attempt part b.ii only.

(c) The diagram shows the levels of typhoid across the world.




(i) Give reasons for the global distribution of typhoid.

**(2)**

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## 3. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

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# 4. Question 8 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

**8** *P. vivax* is one of several parasites that can cause malaria.

(a) Explain how *P. vivax* is transmitted from one person to another.

**(3)**

(b) The table shows the total number of deaths from malaria and the number of deaths caused by *P. vivax*.

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(i) Compare the number of deaths caused by *P. vivax* with the total number of deaths from malaria in regions Q and R.

Include calculations to support your answer.

**(3)**

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# 4. Question 8 · part b.ii · 1 mark

June 2019 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Suggest one reason why the number of deaths from malaria varies across the different regions.

**(1)**

(c) Scientists are developing a vaccine to protect against malaria.

Explain how vaccinating individual people will help to protect a whole population from malaria.

**(3)**

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(d) *P. vivax* reproduces by asexual reproduction and sexual reproduction.

Explain why it is an advantage for a species to reproduce by both methods.

**(3)**

**(Total for Question 8 = 13 marks)**

## 4. Question 8 · part b.ii · 1 mark · Working and answer

Working and answer

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## 5. Question 4 · part c.i · 2 marks

January 2023 Paper 1

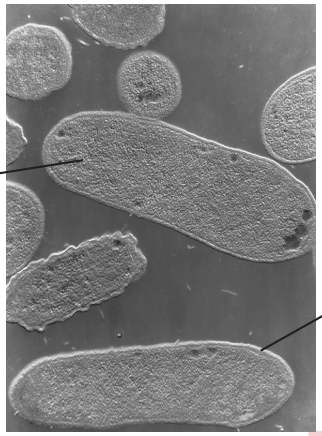
Further practice · Attempt part c.i only.

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(a) The photograph shows *Salmonella typhi*, magnified 30 000 times.




(Source: © CNRI/SCIENCE PHOTO LIBRARY)

(i) Each bacterium is surrounded by cell structure Y.

Identify cell structure Y.

**(1)**

.....

(ii) On the photograph bacterium X has a length of 55 mm.

Calculate the actual length of the bacterium in micrometres (µm).

[1 mm = 1000 µm]

**(2)**

actual length = ..... µm

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# 5. Question 4 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

(b) A team of scientists found that the legs of houseflies have hundreds of different species of bacteria on them.

(i) Flies in towns and cities carry more bacteria than flies that are found in the countryside.

Suggest a reason for this observation.

**(1)**

(ii) One species of bacterium identified is *Helicobacter pylori* which causes stomach ulcers.

Explain how houseflies could increase the transmission of *Helicobacter pylori*.

**(3)**

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## 5. Question 4 · part c.i · 2 marks

January 2023 Paper 1

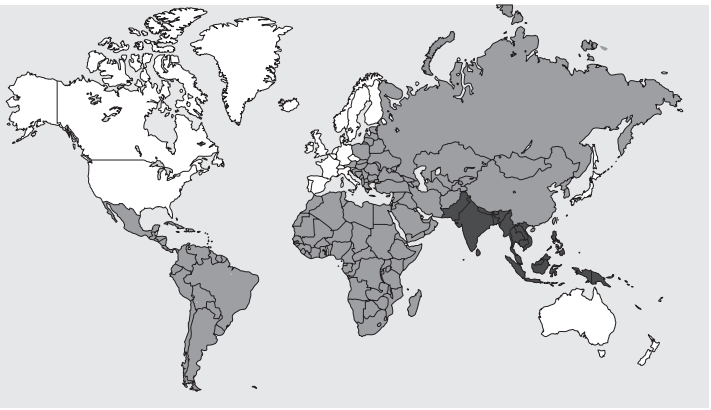
Further practice · Attempt part c.i only.

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(c) The diagram shows the levels of typhoid across the world.




(i) Give reasons for the global distribution of typhoid.

**(2)**

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P 7 2 2 7 1 A 0 1 1 2 4

## 5. Question 4 · part c.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part c.i only.

(ii) Some strains of bacteria that cause typhoid are resistant to antibiotics.

Discuss how resistance to antibiotics develops and the effect this has on the treatment of typhoid.

**(5)**

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**(Total for Question 4 = 14 marks)**

## 5. Question 4 · part c.i · 2 marks · Working and answer

Working and answer

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# Practice workbook · Explain how enzymes work in biosensors and test strips

Glucose strips and sensors using immobilised enzymes behind a partially permeable membrane.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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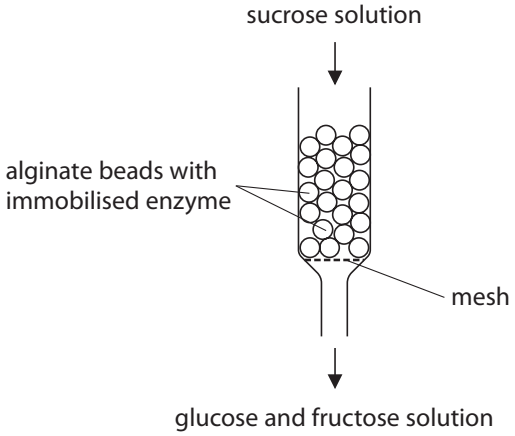
Type ID: HBI-ENZYME-BIOSENSOR-FUNCTION

# 1. Question 6 · part b.ii · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part b.ii only.

**6** (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.




(i) Name the immobilised enzyme used in this process.

**(1)**

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

**(3)**

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.....

(iii) State two other advantages of using immobilised enzymes.

**(2)**

1 .....

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2 .....

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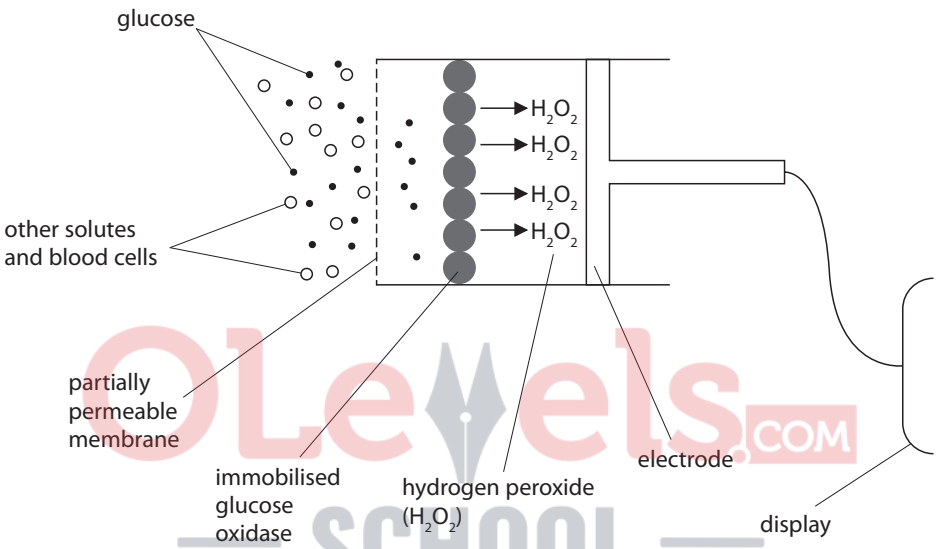
# 1. Question 6 · part b.ii · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part b.ii only.

(b) Biosensors that contain immobilised enzymes can be used to measure the levels of glucose in blood.

The diagram shows this type of biosensor.




(i) Explain the function of the partially permeable membrane.

**(2)**

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# 1. Question 6 · part b.ii · 4 marks

November 2020 Paper 1

Worked-example retry · Attempt part b.ii only.

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(ii) Explain the function of the immobilised enzyme in the biosensor.

Use information from the diagram to help your answer.

**(4)**

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**(Total for Question 6 = 12 marks)**

## 1. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

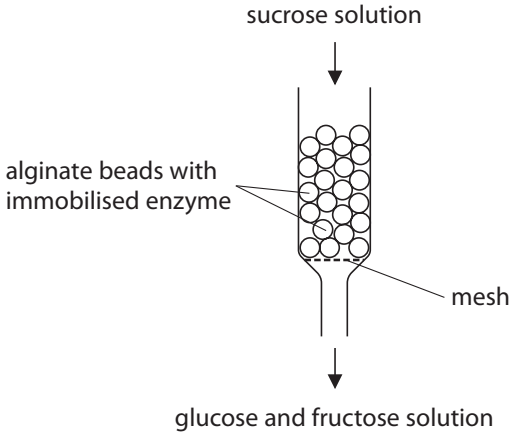
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## 2. Question 6 · part b.i · 2 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

**6** (a) The diagram shows how immobilised enzymes can be used to convert sucrose into glucose and fructose.




(i) Name the immobilised enzyme used in this process.

**(1)**

.....

(ii) One advantage of using immobilised enzymes is that they do not contaminate the products, glucose and fructose.

Explain how the products could be tested to show that they had not been contaminated by the enzyme.

**(3)**

.....

.....

.....

.....

.....

.....

(iii) State two other advantages of using immobilised enzymes.

**(2)**

1 .....

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2 .....

.....

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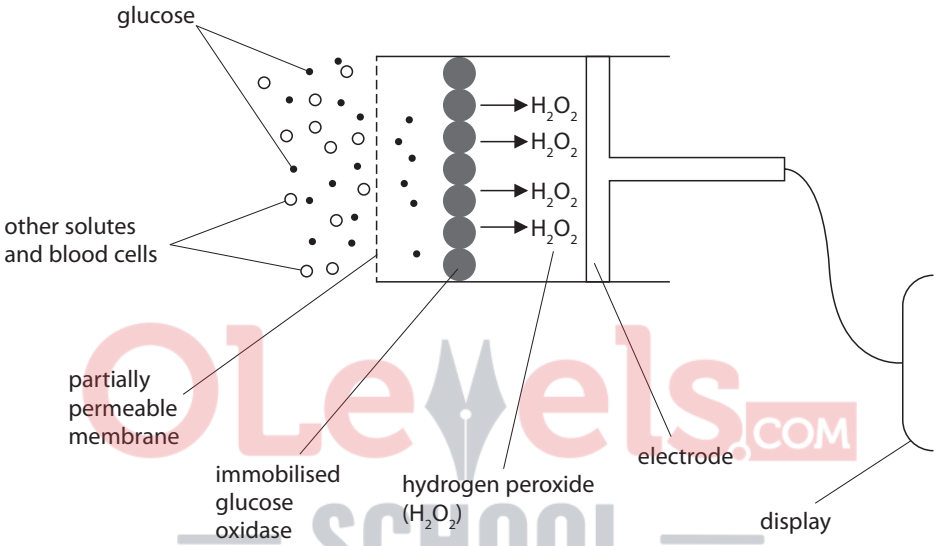
## 2. Question 6 · part b.i · 2 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

(b) Biosensors that contain immobilised enzymes can be used to measure the levels of glucose in blood.

The diagram shows this type of biosensor.




(i) Explain the function of the partially permeable membrane.

**(2)**

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## 2. Question 6 · part b.i · 2 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

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(ii) Explain the function of the immobilised enzyme in the biosensor.

Use information from the diagram to help your answer.

**(4)**

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**(Total for Question 6 = 12 marks)**

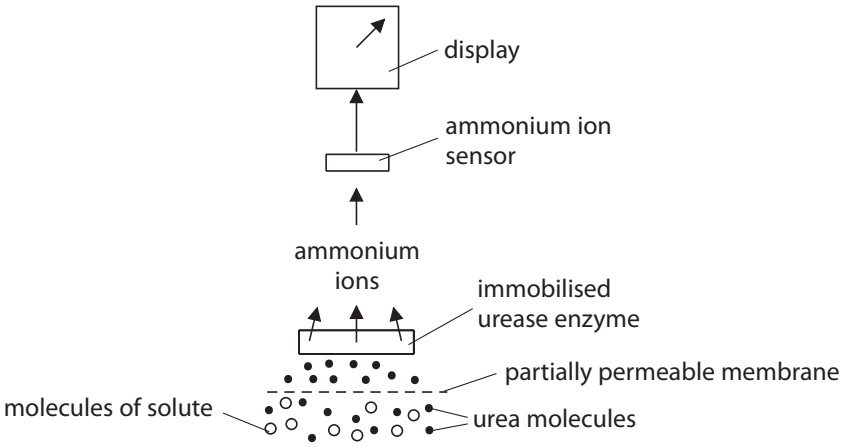
Working and answer

# 3. Question 3 · part a.ii · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.ii only.

**3** The diagram shows a biosensor that can be used to measure the concentration of urea in urine.




(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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## 3. Question 3 · part a.ii · 2 marks · Working and answer

Working and answer

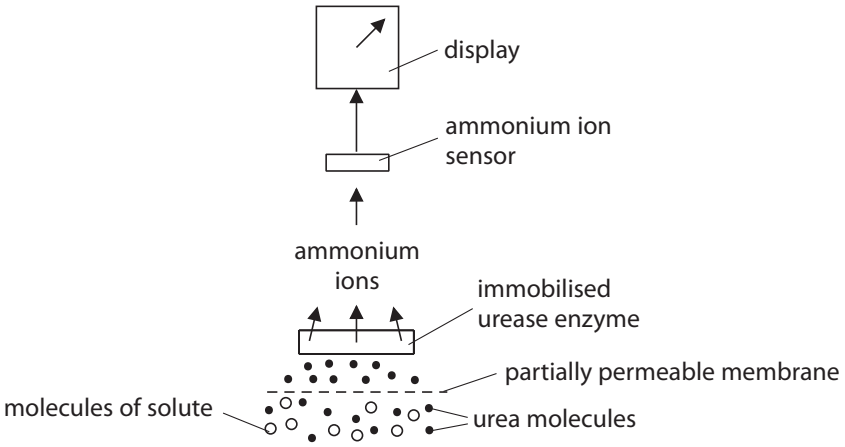
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# 4. Question 3 · part a.iii · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.iii only.

**3** The diagram shows a biosensor that can be used to measure the concentration of urea in urine.




(a) (i) The urea molecules are in solution.

State the name of the solvent. **(1)**

(ii) Explain the function of the partially-permeable membrane. **(2)**

(iii) Suggest the function of the urease enzyme. **(3)**

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## 4. Question 3 · part a.iii · 3 marks

June 2021 Paper 2

Further practice · Attempt part a.iii only.

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(b) A scientist wants to use the biosensor to compare the urea concentration of two urine samples.

Explain why it is important to do the tests at the same temperature.

**(4)**

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(c) Immobilised enzymes are also used in industry to produce useful molecules.

Give two advantages of using immobilised enzymes instead of enzymes that are free to move in solution.

**(2)**

1

2

**(Total for Question 3 = 12 marks)**

## 4. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

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## 5. Question 7 · part b · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b only.

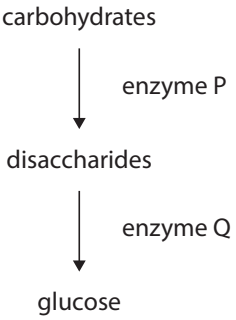
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**7** (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.




Name enzyme P.

**(1)**

.....

(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

**(2)**

.....

.....

.....

.....

(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

**(1)**

.....

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## 5. Question 7 · part b · 2 marks · Working and answer

Working and answer

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# Practice workbook · Explain how the pupil responds to changes in light intensity

Iris muscle actions that widen or narrow the pupil in dim or bright light and why a quick response protects the retina.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-EXPLAIN-PUPIL-LIGHT-RESPONSE

# 1. Question 6 · part b.i · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b.i only.

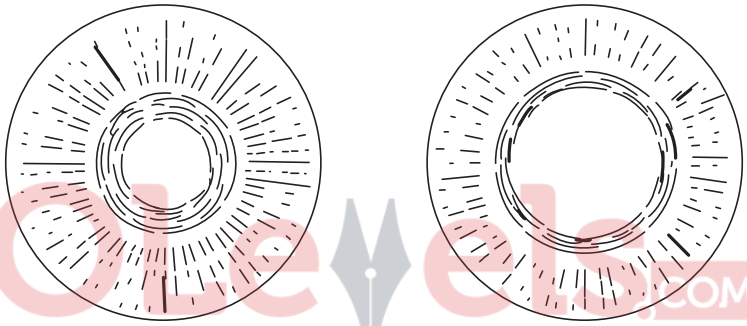
**6** The human eye consists of many parts.

(a) Which part has the highest concentration of light sensitive cells?

**(1)**

- ☒ **A** blind spot
- ☒ **B** choroid
- ☒ **C** fovea
- ☒ **D** optic nerve

(b) The diagram shows the iris of an eye in two different light conditions, A and B.



light condition A



light condition B

(i) Describe what happens in the eye when the light condition changes from A to B.

**(5)**

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P 6 6 4 3 3 A 0 1 4 2 4

# 1. Question 6 · part b.i · 5 marks

June 2021 Paper 2

Worked-example retry · Attempt part b.i only.

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(ii) Explain why it is important that the changes in the eye occur quickly when the light condition changes from B to A.

**(3)**

**(Total for Question 6 = 9 marks)**

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Working and answer

# 2. Question 6 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

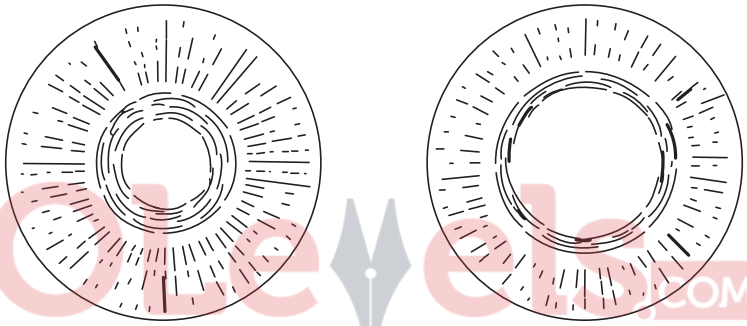
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(a) Which part has the highest concentration of light sensitive cells?

**(1)**

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- ☒ **D** optic nerve

(b) The diagram shows the iris of an eye in two different light conditions, A and B.



light condition A light condition B

(i) Describe what happens in the eye when the light condition changes from A to B.

**(5)**

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## 2. Question 6 · part b.ii · 3 marks

June 2021 Paper 2

Further practice · Attempt part b.ii only.

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(ii) Explain why it is important that the changes in the eye occur quickly when the light condition changes from B to A.

**(3)**

**(Total for Question 6 = 9 marks)**

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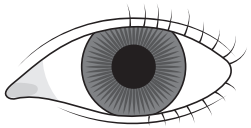
Working and answer

## 3. Question 6 · part c.i · 3 marks

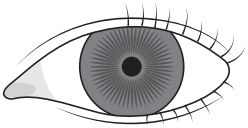
June 2024 Paper 2

Further practice · Attempt part c.i only.

**6** The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

**(3)**

(b) A student calculated the area of the pupil in bright light to be 7.1 $mm^2$.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = $\pi r^2$ $\pi = 3.14$]

**(3)**

area of pupil in dim light = ..... $mm^2$

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P 7 4 6 3 0 A 0 1 6 2 4

## 3. Question 6 · part c.i · 3 marks

June 2024 Paper 2

Further practice · Attempt part c.i only.

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(c) (i) Explain how the size of the pupil is adjusted from bright light to dim light.

**(3)**

(ii) Explain the adjustment made to the pupil when moving into dim light.

**(2)**

**(Total for Question 6 = 11 marks)**

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## 3. Question 6 · part c.i · 3 marks · Working and answer

Working and answer

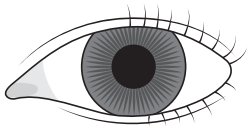
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## 4. Question 6 · part c.ii · 2 marks

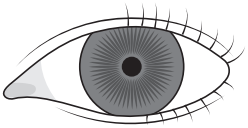
June 2024 Paper 2

Further practice · Attempt part c.ii only.

**6** The diagram shows how the size of the pupil changes in different light intensities.



Dim light



Bright light

(a) Describe a method that could be used to investigate how the size of the pupil changes in different light intensities.

**(3)**

(b) A student calculated the area of the pupil in bright light to be 7.1 $mm^2$.

Using information from the diagram, calculate the area of the pupil in dim light.

[for a circle, area = $\pi r^2$ $\pi = 3.14$]

**(3)**

area of pupil in dim light = ..... $mm^2$

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# 4. Question 6 · part c.ii · 2 marks

June 2024 Paper 2

Further practice · Attempt part c.ii only.

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(c) (i) Explain how the size of the pupil is adjusted from bright light to dim light.

**(3)**

(ii) Explain the adjustment made to the pupil when moving into dim light.

**(2)**

**(Total for Question 6 = 11 marks)**

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Working and answer

# Practice workbook · Interpret a menstrual-cycle graph of hormones or lining thickness

Graphs of hormone concentration or uterine lining width across a 28-day cycle.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-INTERPRET-MENSTRUAL-CYCLE-GRAPH

# 1. Question 4 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

**4** (a) State three methods of birth control.

**(3)**

1 .....

.....

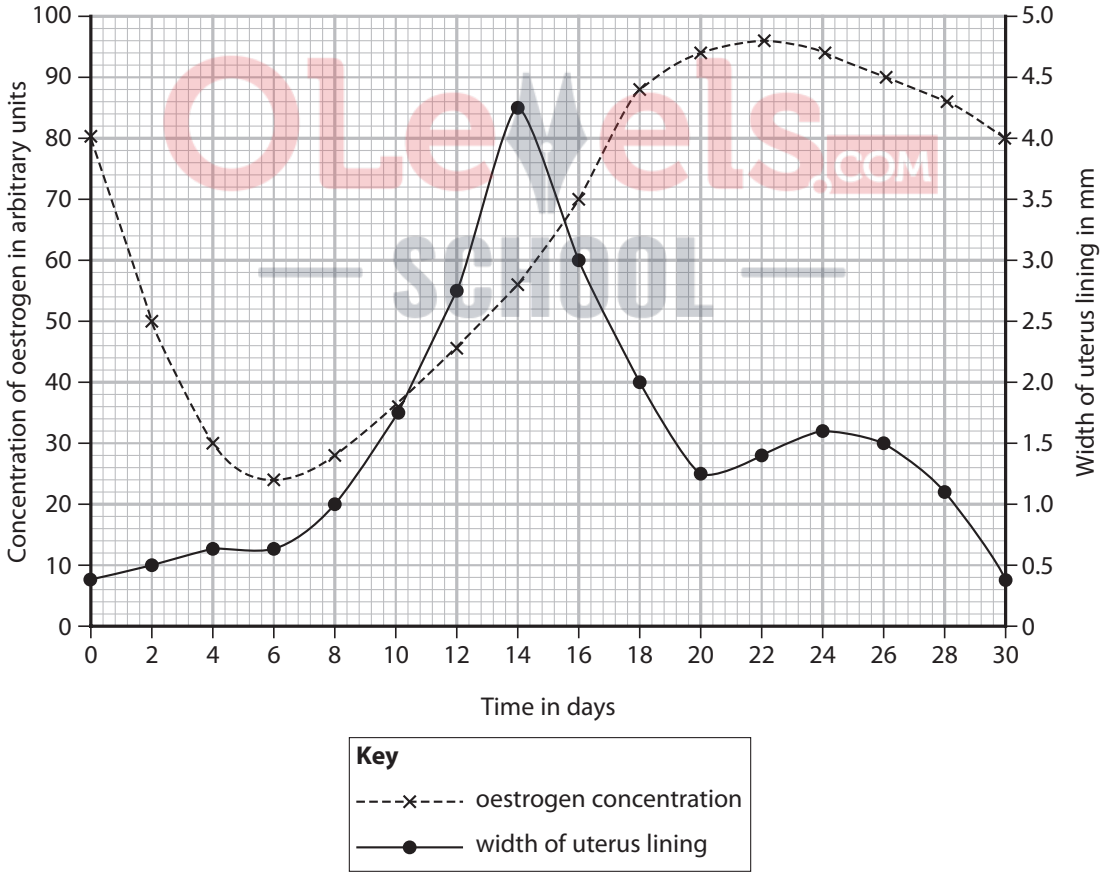
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(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.




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# 1. Question 4 · part b.ii · 3 marks

June 2022 Paper 2R

Worked-example retry · Attempt part b.ii only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

**(2)**

change in width = ..... mm

(ii) Explain the importance of this change in width.

**(3)**

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(iii) Determine during which days menstruation occurred.

**(1)**

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

**(3)**

**(Total for Question 4 = 12 marks)**

## 1. Question 4 · part b.ii · 3 marks · Working and answer

Working and answer

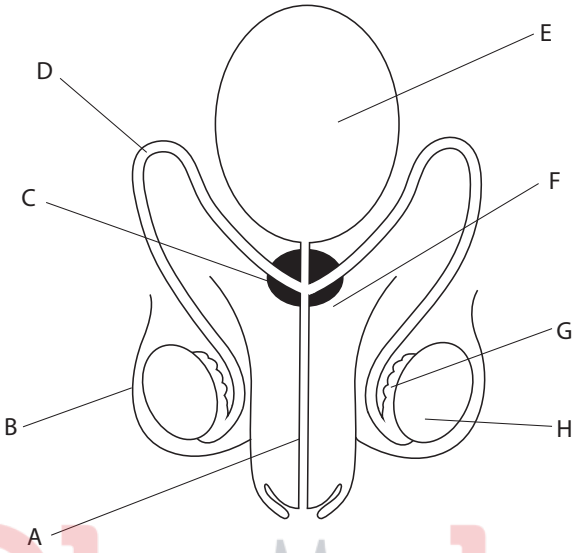
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## 2. Question 7 · part b.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.i only.

**7** (a) The diagram shows the male reproductive system.




The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

**(4)**

| Description of the structure | Letter |
|---|---|
| stores sperm | |
| produces sperm | |
| produces fluid to help sperm move | |
| contains cells dividing by meiosis | |

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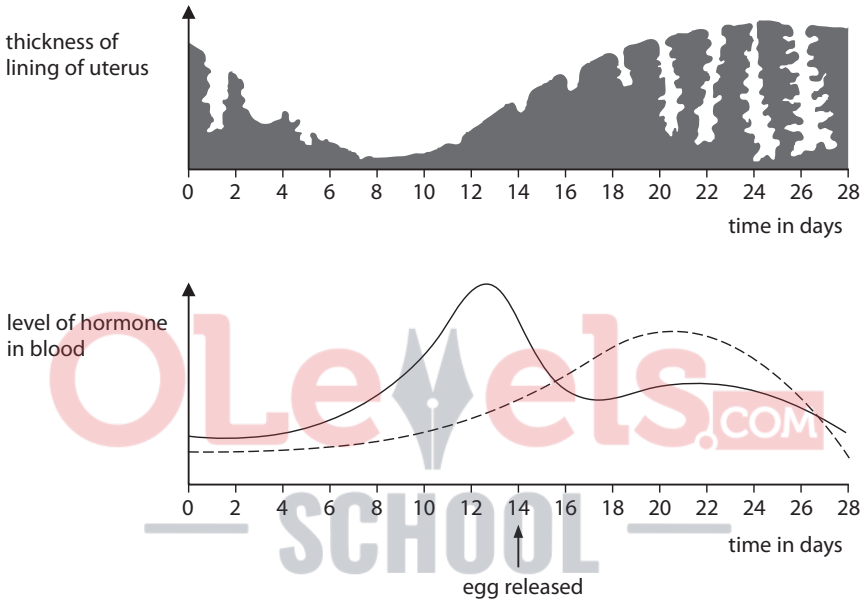
## 2. Question 7 · part b.i · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.i only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.




(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

**(2)**

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

**(2)**

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## 2. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

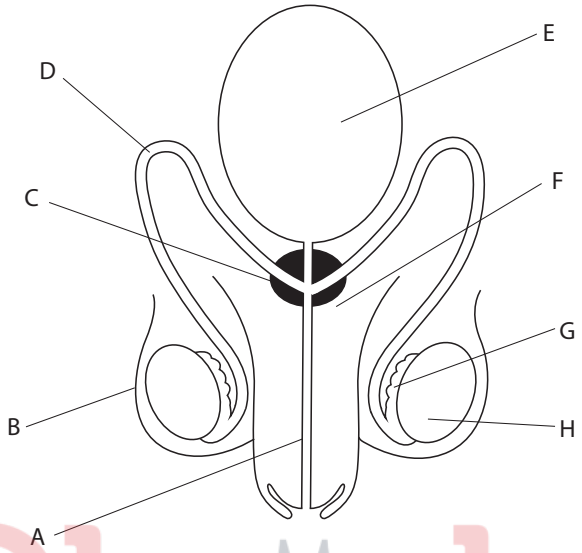
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## 3. Question 7 · part b.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

**7** (a) The diagram shows the male reproductive system.




The table gives descriptions of structures in the male reproductive system.

Complete the table by giving a letter from the diagram to match each description.

Each letter may be used once, more than once or not at all.

**(4)**

| Description of the structure | Letter |
|---|---|
| stores sperm | |
| produces sperm | |
| produces fluid to help sperm move | |
| contains cells dividing by meiosis | |

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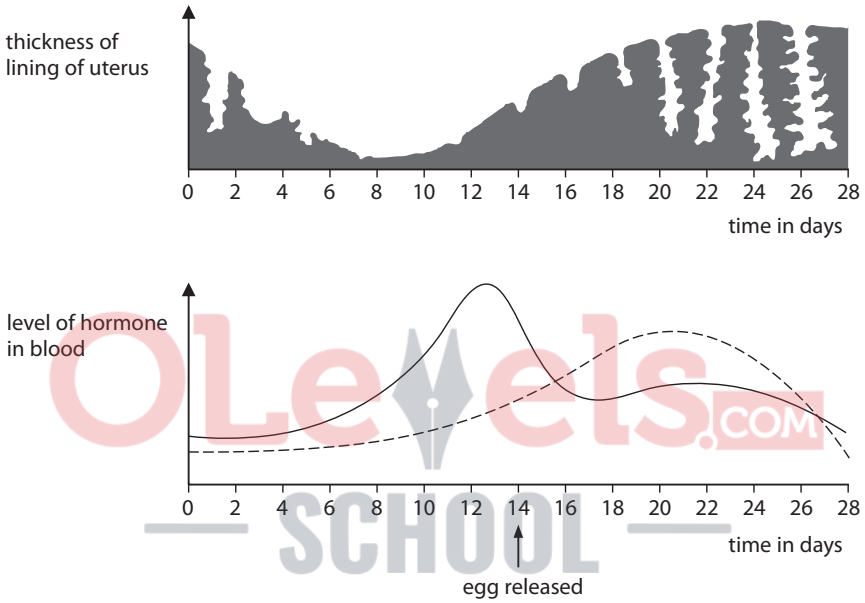
# 3. Question 7 · part b.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(b) Changes occur in the female body during the menstrual cycle.

The graphs show changes in the lining of the uterus and the levels of two hormones that control these changes.




(i) Use information from the graphs to explain the cause of growth in the lining of the uterus after day 12.

**(2)**

(ii) Explain why fertilisation is less likely to occur from intercourse between days 8 and 12.

**(2)**

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## 3. Question 7 · part b.ii · 2 marks

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(c) A synthetic oestrogen called oestradiol is used to make contraceptive pills. Natural oestrogen, which is a lipid molecule, could be used but is less effective than oestradiol when taken as a pill.

(i) Suggest why natural oestrogen is not as effective as oestradiol when taken as a pill.
**(2)**

(ii) Give a reason why contraceptive pills only need to be taken for 21 days out of the 28-day cycle.
**(1)**

**(Total for Question 7 = 11 marks)**

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Working and answer

# 4. Question 4 · part b.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.iii only.

**4** (a) State three methods of birth control.

**(3)**

1 .....

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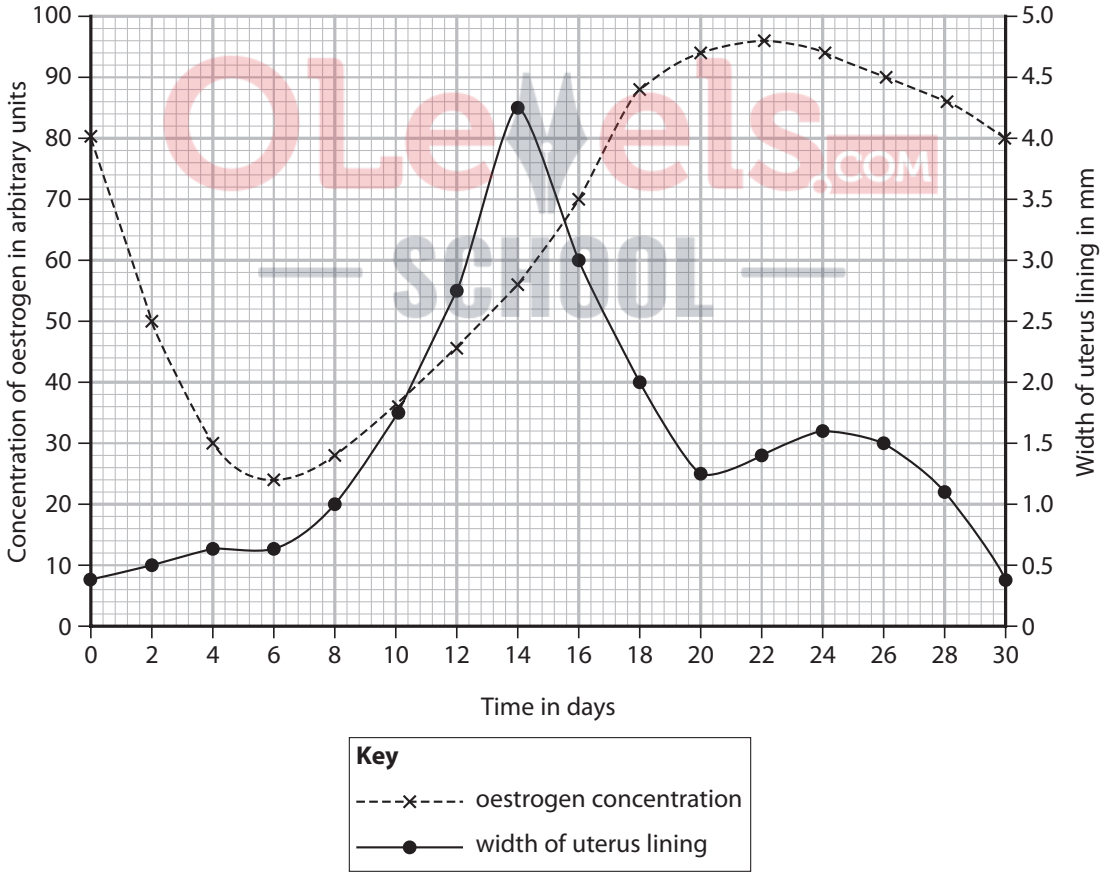
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3 .....

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(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.




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# 4. Question 4 · part b.iii · 1 mark

June 2022 Paper 2R

Further practice · Attempt part b.iii only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

**(2)**

change in width = ..... mm

(ii) Explain the importance of this change in width.

**(3)**

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(iii) Determine during which days menstruation occurred.

**(1)**

.....

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

**(3)**

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**(Total for Question 4 = 12 marks)**

## 4. Question 4 · part b.iii · 1 mark · Working and answer

Working and answer

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## 5. Question 4 · part b.iv · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.iv only.

**4** (a) State three methods of birth control.

**(3)**

1 .....

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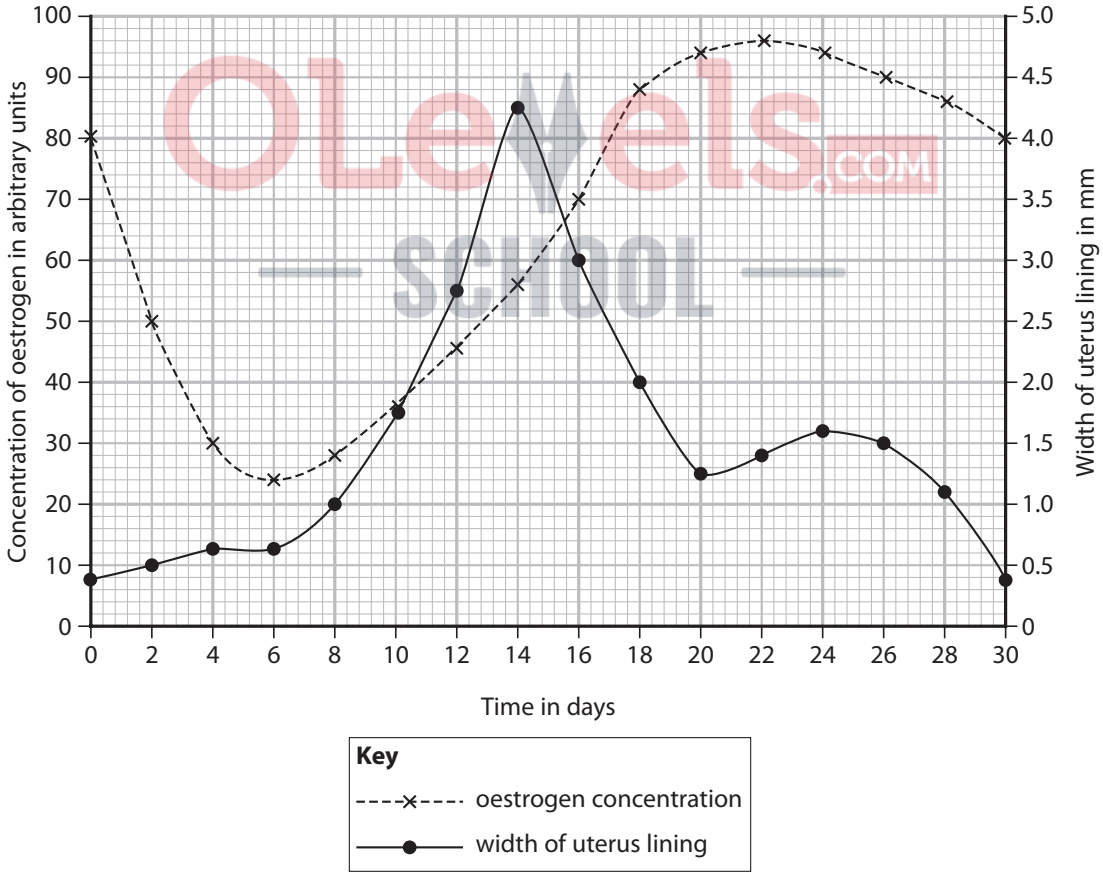
2 .....

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3 .....

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(b) The graph shows the changes in the width of the uterus lining and in the concentration of oestrogen in the blood during one menstrual cycle.




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# 5. Question 4 · part b.iv · 3 marks

June 2022 Paper 2R

Further practice · Attempt part b.iv only.

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(i) Determine the change in width of the uterus lining between day 6 and day 14.

**(2)**

change in width = ..... mm

(ii) Explain the importance of this change in width.

**(3)**

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(iii) Determine during which days menstruation occurred.

**(1)**

.....

(iv) Ovulation takes place on day 15.

Describe how the two lines on the graph would be different if the ovum had been fertilised.

**(3)**

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**(Total for Question 4 = 12 marks)**

## 5. Question 4 · part b.iv · 3 marks · Working and answer

Working and answer

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